
THREE-DIMENSIONAL KINEMATIC ANALYSIS OF THE SNATCH TECHNIQUE FOR LIFTING DIFFERENT BARBELL WEIGHTS

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ABSTRACT

Hadi, G, Akkuş, H, and Harbili, E. Three-dimensional kinematic analysis of snatch technique for lifting different barbell weights. *J Strength Cond Res* 26(6): 1568–1576, 2012—The purpose of this study was to investigate the effects of increased barbell loads on barbell and body kinematics of the snatch lifts at 60, 80, and 100% of 1 repetition maximum and to evaluate the biomechanics of snatch technique. The study was performed on 7 elite male weightlifters of the Turkish national team. Four cameras operating at 50 fields per second were used to record the lifts. For 3D kinematic analysis of center of gravity (CG) and barbell movement, the points on the body and the barbell were digitized by using an Ariel Performance Analysis System. There were significant differences between the vertical work values ($p < 0.05$). The power values of the 3 snatch lifts were also found to be significantly different ($p < 0.05$). Another significant difference ($p < 0.05$) was observed between maximum vertical displacement of the barbell, maximum vertical velocity of the barbell, maximum vertical displacement of CG, the vertical velocity of CG during the turnover under the barbell. The results demonstrated that vertical and horizontal kinematics of the barbell and body decreased at the pull phase of the snatch technique as the barbell load increased. The power output during the second pull increased although the work done did not change, whereas work and power output increased during the first pull phase depending on the increase in the barbell weight. The finding of this study suggested that weightlifters had to perform the turnover under the barbell and the catch phase faster, because when the barbell weight was increased at snatch lift, vertical kinematics of the barbell decreased.

KEY WORDS mechanical work, power output, center of gravity

INTRODUCTION

The aim of Olympic weightlifting competitions is always to lift higher weights successfully. Lifting higher weights in snatch lifts requires a multifactor performance including technique, power, explosive strength, and flexibility (2,5,9,10,15–18,22). When the snatch technique is analyzed as a whole, it can be seen that the synchronization and perfection of the system consisting of the body and barbell is the key to a successful lift (2,5). The performance pattern of the snatch technique requires the barbell to be lifted from the floor to a straight-arm overhead position in one continuous movement (2,5,15–18). The first 5 phases (first pull, transition, second pull, turnover under the barbell, and catch phase) are considered to be the most important phases of the snatch lifts, and increasing the barbell weight has an important effect on all biomechanical factors during these phases (15–18). Determining the exact effects of the increased barbell weight on the barbell and body kinematics might help to understand the effective technical factors and the biomechanics of successful lifts of higher weights.

A great majority of the published studies on the biomechanics of weightlifting have usually focused on the kinematics of the barbell and body segments of elite weightlifters who participated in national (1,7,8,10,17,19,24,25) and international competitions (3,4,6,9) and world championships (12). The common aim of those studies was to determine the biomechanical differences between the different phases and to evaluate the technical components of snatch lifts. These studies reported that the mechanical work during the first pull was higher than that of the second pull and that the vertical velocity of the barbell reached maximum levels during the second pull (17). On the other hand, it was expressed in a large number of studies that mechanical power was higher in the second pull compared with that in the first pull (2,5,9,12). The mechanical work and power outputs showed that the first pull was characterized by force, whereas the second pull was largely of power nature (18). Therefore, the higher power output seen during the second pull was reported to be a result of the combination of a greater velocity of the barbell and a shorter duration during the second pull (13). Isaka et al. (21) stated that optimizing the barbell height after the second pull and minimizing the decrease after the maximum height of the barbell were effective factors for a successful lift. Similar

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results were reported in a study by Burdett (3) showing that lifting the barbell to the lowest possible height by talented weightlifters created a certain advantage, and minimizing the decrease of the barbell height during the turnover under the barbell resulted in savings at total work. The horizontal displacement pattern of the barbell during the lift is also important (25) in that the horizontal movement during the first pull and transition phase allows the storage of elastic energy into the extensor muscles of the knees during the flexion of the knees and its use during the following concentric contraction of the knees, resulting in an explosive power output during the second pull (16). The horizontal displacement of the barbell is a toward-away-toward pattern: The barbell is pulled toward the body during the first pull and the transition phase, and, during the second pull, it moves away from the lifter's body, and finally, it drops toward the body from the maximum height (16,20). Stone et al. (25) reported that the barbell was displaced horizontally by 10–20 cm during the snatch lifts by elite weightlifters. However, these anterior-posterior displacements of the barbell should be small to avoid unnecessary energy consumption (16).

There were relatively a limited number of studies available in literature examining the effects of the increased barbell weight on the biomechanics of the barbell and body (12). In their study, Garhammer (12) found that a small decrease in weight, about 5%, could often increase power output substantially because of a considerably greater movement speed and shorter time interval for the completion of the lift.

In the light of these findings, the kinematic analysis of the body and barbell during the snatch technique may provide important information to increase performance by determining the mechanical factors that are effective in lifting higher weights. Increasing the barbell weight results in a decrease in power output, vertical velocity, and maximum height of the barbell. A detailed analysis of the kinematics of the increased barbell weight may provide useful information for coaches and weightlifters, who always aim to lift higher weights.

The purpose of this study was to investigate the effects of increased barbell weights on the barbell and body kinematics of elite weightlifters at different percentages (60, 80, and 100%) of 1 repetition maximum (1RM) in the snatch lifts and to evaluate the biomechanics of the snatch technique.

METHODS

Experimental Approach to the Problem

To determine the changes that the increased barbell weight caused on the body and barbell kinematics, the kinematic data obtained from the subjects were studied by 3-dimensional biomechanical analysis. The reason for the preference for the 3-dimensional analysis was that other studies reported that 2-dimensional analysis adversely affected the reliability of angular kinematics (2).

Subjects

This study was performed on 7 elite male weightlifters (Table 1) at different categories of the Turkish weightlifting national team during their preparation camp for the 2007 European Championship. The subjects were medal-winning weightlifters in previous international competitions. All the subjects provided written informed consent approved by the local ethics committee of the Selcuk University.

Procedures

To determine the 3-dimensional kinematic data of the barbell and body segments during the snatch technique and to digitize the data provided with higher precision, a set of 14 reflective markers were stamped on tiptoe, ankle, knee, hip, shoulder, elbow, and wrist on both sides of the body and 2 markers on the chin and forehead. The other 2 markers were placed at the right and left ends of the barbell. The reflective markers on the body and the barbell were digitized by using an Ariel Performance Analysis System (San Diego, CA, USA). To obtain a clearer vision of the markers, an illumination source of 500 W of power was placed on each camera (Figure 1).

TABLE 1. Characteristics of the subjects and the barbell.

Subjects	Age (y)	Body mass (kg)	Height (cm)	Category (kg)	Barbell mass (kg)				
					60%1RM	80%1RM	100%1RM	WR*	WR (%)
GA	27	58	150	56	66	88	110	143	78
SA	31	60	162	56	72	96	120	138	87
EB	20	64	156	62	72	96	120	153	78
YA	29	70	163	69	84	112	140	165	85
TS	22	81	171	77	78	104	130	174	75
HY	25	100	175	94	96	128	160	188	85
BS	32	107	176	105	96	128	160	200	80

*World record.

†the ratio between 100%1RM and WR.

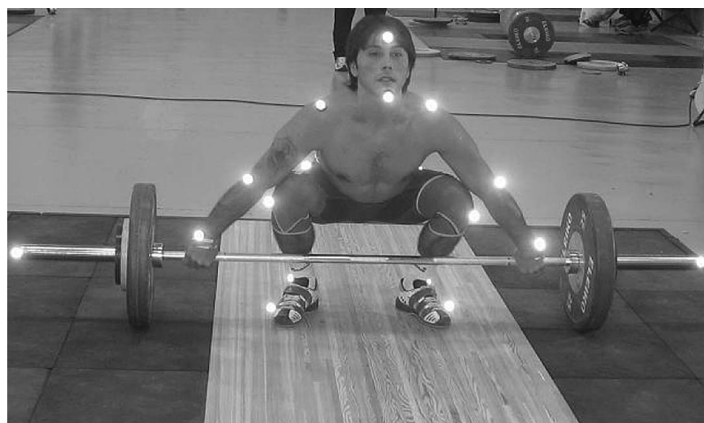


Figure 1. Placement points of reflective markers on the body and barbell.

Four digital cameras (Sony DCR-TRV18E, Tokyo, Japan) capturing 50 fields per second with approximately 3 minutes of rest between lifts were positioned on the diagonal level of the platform at a distance of 7 m away from the platform forming an angle of about 45° with the sagittal plane of the weightlifter. Cameras 1 and 2 were placed to view the right side of the barbell and the body, and cameras 3 and 4 recorded the left side. There was an angle of about 90° between the optical axes (Figure 2).

The cameras were synchronized using the lift-off of the barbell, and 3-dimensional coordinates were constructed using the direct linear transformation method. To calibrate the viewing area in 3 dimensions, a rectangular cube with 12 control points of 250-cm length, 100-cm breadth, and 200-cm height was used. The video recordings were taken at the fourth week of the 4-month preparation camp before the 2007 European

Championship. Only successful lifts of each weightlifter at 60% (60%1RM), 80% (80%1RM), and 100% of 1RM (100%1RM) were recorded. A low-pass digital filter with a cutoff frequency of 4 Hz was used for the smoothing of the raw data (15,16,18).

The snatch technique was divided into 5 phases according to the changes in the knee angle and the vertical position of the barbell as the first pull, the transition, the second pull, the turnover under the barbell, and the catch phase (Figure 3).

The linear kinematics of the body and barbell and the angular displacements of the knee joint were calculated. The vertical

work done on the barbell during the first pull and the second pull was calculated from the change in mechanical energy, which is defined as the sum of kinetic and potential energy (11). The work done in elevating the center of gravity (CG) of the body was calculated from potential energy, and the horizontal work done on the barbell was calculated by multiplying the horizontal displacement of the barbell by the horizontal force, obtained from the multiplication of horizontal acceleration of the barbell by its mass. The power output of weightlifters during the first and second pulls was calculated by dividing the work of each phase by its duration. Technical efficiency was calculated by dividing the vertical work by the total work on the barbell. All of the work, power values, and technical efficiency were calculated according to the methodology described by Garhammer (11).

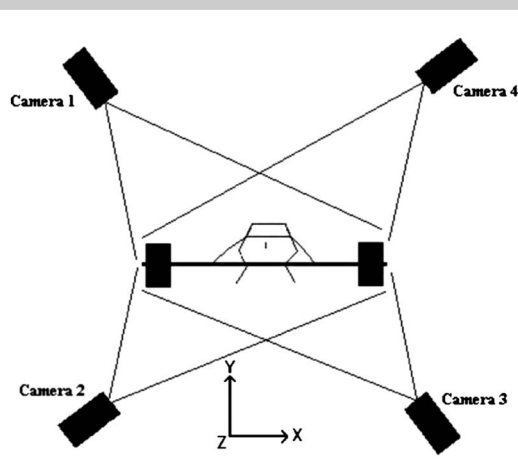


Figure 2. Top view of the positions of the cameras recording snatch lifts.

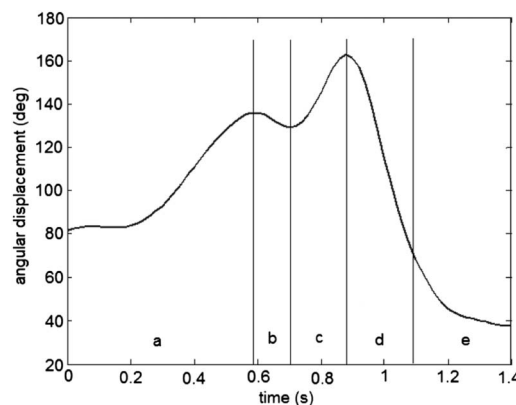


Figure 3. Phases of the snatch technique based on the angular displacement at knee angle. (a) the first pull, (b) the transition, (c) the second pull, (d) the turnover under the barbell, and (e) the catch.

TABLE 2. Duration (seconds) of the phases in snatch lifts.*

	Mean $\pm$ SD			Wilks' lambda
	60%1RM	80%1RM	100%1RM	
First pull (s)	0.42 $\pm$ 0.06	0.40 $\pm$ 0.03‡	0.46 $\pm$ 0.04	0.149
Transition (s)	0.10 $\pm$ 0.01	0.11 $\pm$ 0.01	0.13 $\pm$ 0.03	0.591
Second pull (s)	0.14 $\pm$ 0.03	0.13 $\pm$ 0.01	0.12 $\pm$ 0.01	0.524
Turnover under the barbell (s)	0.24 $\pm$ 0.03	0.24 $\pm$ 0.03	0.23 $\pm$ 0.02	0.880

*1RM = 1 repetition maximum.

†Significant difference at the $p < 0.05$ level in 80%1RM.

‡Significant difference at the $p < 0.05$ level in 100%1RM.

§Significant difference at the $p < 0.05$ level in 60%1RM.

|| $p < 0.05$.

Statistical Analyses

Multivariate test Wilks' lambda (λ) and post hoc test with Bonferroni correction were used for the statistical analysis of the linear kinematics and energetics of the barbell. Normality of data was verified by the Kolmogorov-Smirnov test. The level of statistical significance was set at $p \leq 0.05$.

RESULTS

No significant differences were found between the durations of the phases, except for the first pull (Table 2).

It was indicated that there were significant differences between the maximum height of the barbell during 60%1RM, 80%1RM, and 100%1RM snatch lifts (Figure 4) and between the maximum vertical velocity of the barbell (Figure 5). The maximum height and vertical velocity of the barbell decreased significantly during 60%1RM, 80%1RM, and 100%1RM snatch lifts, respectively. Significant differences were also found in the maximum vertical

displacement and vertical velocity of CG during the turnover under the barbell. Another significant difference was observed in the drop distance of the barbell from maximum height (Table 3).

It was found that there were significant differences between the percentages of 1RM in vertical work done during the first pull and in the total vertical work. The vertical work in the first pull and the total vertical work were significantly greater at 100%1RM, 80%1RM, and 60%1RM snatch lifts, respectively. Another significant difference was found between 80%1RM and 100%1RM in the total vertical work done in elevating CG during the lifts. There was also a significant difference between the vertical work values of CG in the second pull. In addition, the technical efficiency value was found to be significantly greater at 100%1RM than at 60%1RM. However, no significant differences were found between the percentages of 1RM in the total horizontal work or the vertical or horizontal work in the second pull (Table 4).

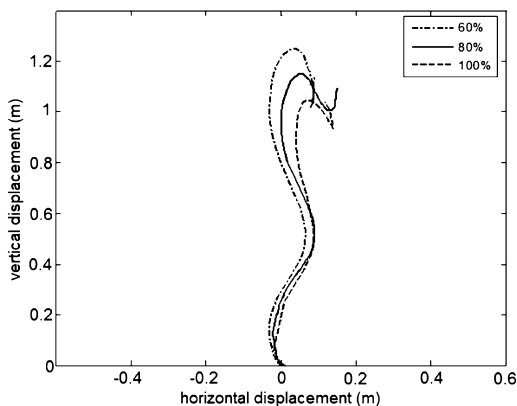


Figure 4. Trajectories of the barbell in the snatch lifts of TS: from lift-off to the catch phase.

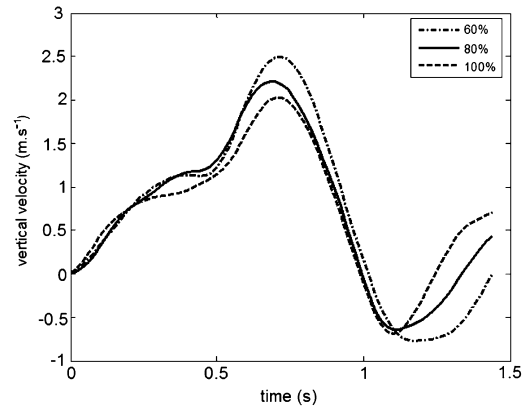


Figure 5. The linear vertical velocity of the barbell in the snatch lifts of TS: from lift-off to the catch phase.

TABLE 3. Kinematics of the barbell and CG during snatch lifts.*

	Mean $\pm$ SD			Wilks' lambda
	60%1RM	80%1RM	100%1RM	
Maximum height of the barbell (m)	1.38 $\pm$ 0.09 [†]	1.29 $\pm$ 0.08 [‡]	1.18 $\pm$ 0.09 [§]	0.027
Maximum vertical displacement of CG (m)	0.45 $\pm$ 0.04	0.44 $\pm$ 0.03	0.43 $\pm$ 0.03 [§]	0.118
Maximum vertical velocity of the barbell (m·s ⁻¹)	2.30 $\pm$ 0.18 [†]	2.02 $\pm$ 0.18 [‡]	1.79 $\pm$ 0.17 [§]	0.030
Vertical velocity of CG during the turnover under the barbell (m·s ⁻¹)	-1.49 $\pm$ 0.16 [†]	-1.73 $\pm$ 0.14 [‡]	-1.97 $\pm$ 0.13 [§]	0.077
Drop distance of the barbell (m)	0.24 $\pm$ 0.07 [†]	0.16 $\pm$ 0.04	0.13 $\pm$ 0.03 [§]	0.107

*1RM = 1 repetition maximum.

[†]Significant difference at the $p < 0.05$ level in 80%1RM.[‡]Significant difference at the $p < 0.05$ level in 100%1RM.[§]Significant difference at the $p < 0.05$ level in 60%1RM.|| $p < 0.05$.

It was found that there were significant differences between the percentages of 1RM in the vertical power of both the first and the second pulls. The total vertical power and the total horizontal power values of 60%1RM, 80%1RM, and 100%1RM were also significantly different. Although the multivariate test showed no significant differences between the values of the horizontal power in the second pull ($\lambda = 0.304$; $p = 0.051$), the Bonferroni test showed that the horizontal power in the second pull for 80%1RM was significantly higher than that of 100%1RM ($p < 0.05$).

Another significant difference was observed in the total vertical power values of CG. On the other hand, there was no significant difference between the vertical power values of CG during the second pull ($\lambda = 0.326$; $p = 0.061$); nevertheless, as far as the results obtained by Bonferroni test are concerned, the vertical power of CG in the second pull of 100%1RM was significantly lower than that of 60%1RM ($p < 0.05$). It was also found that there were no significant differences between the absolute total power values ($\lambda = 0.345$; $p = 0.070$), but the Bonferroni test results

TABLE 4. Work done by CG, horizontal and vertical work done on the barbell during snatch lifts.*

	60%1RM	80%1RM	100%1RM	Wilks' lambda
	Mean $\pm$ SD	Mean $\pm$ SD	Mean $\pm$ SD	
Vertical work in the first pull (J)	389.00 $\pm$ 127.98 [†]	503.00 $\pm$ 154.34 [‡]	630.85 $\pm$ 180.72 [§]	0.049
Vertical work in the second pull (J)	446.42 $\pm$ 78.02	467.00 $\pm$ 84.26	511.14 $\pm$ 97.72	0.529
Total vertical work (J)	835.42 $\pm$ 151.13 [†]	1,014.28 $\pm$ 182.93 [‡]	1,142.14 $\pm$ 217.33 [§]	0.054
Total horizontal work (J)	50.71 $\pm$ 26.71	55.28 $\pm$ 40.47	39.28 $\pm$ 32.04	0.354
Horizontal work in the second pull (J)	38.14 $\pm$ 21.65	44.00 $\pm$ 33.11	22.85 $\pm$ 18.25	0.344
Total vertical work of CG (J)	338.85 $\pm$ 119.73	337.85 $\pm$ 109.13 [‡]	326.14 $\pm$ 104.44	0.256
Vertical work of CG in the second pull (J)	140.42 $\pm$ 36.60	132.00 $\pm$ 41.88	108.85 $\pm$ 25.39 [§]	0.209
Efficiency (%)	94.57 $\pm$ 1.98	95.28 $\pm$ 2.81	97.00 $\pm$ 2.00 [§]	0.161

*CG = center of gravity; efficiency = vertical work/total work on barbell; 1RM = 1 repetition maximum.

[†]Significant difference at the $p < 0.05$ level in 80%1RM.[‡]Significant difference at the $p < 0.05$ level in 100%1RM.[§]Significant difference at the $p < 0.05$ level in 60%1RM.|| $p < 0.05$.

TABLE 5. Power output during snatch lifts.*

	60%1RM	80%1RM	100%1RM	
	Mean $\pm$ SD	Mean $\pm$ SD	Mean $\pm$ SD	Wilks' lambda
Vertical power in the first pull (W)	817.71 $\pm$ 239.49†	1,008.42 $\pm$ 231.76	1,081.28 $\pm$ 199.27‡	0.115
Vertical power in the second pull (W)	2,026.71 $\pm$ 443.31	2,309.00 $\pm$ 462.55	2,595.28 $\pm$ 569.45‡	0.229
Total vertical power (W)	1,174.28 $\pm$ 231.36†	1,413.85 $\pm$ 239.20	1,467.28 $\pm$ 241.09‡	0.131
Total horizontal power (W)	71.14 $\pm$ 35.88	76.42 $\pm$ 54.79§	48.71 $\pm$ 40.40	0.204
Horizontal power in the second pull (W)	170.42 $\pm$ 89.89	200.57 $\pm$ 155.84§	126.28 $\pm$ 118.70	0.304¶
Total vertical power of CG (W)	477.14 $\pm$ 182.15	469.14 $\pm$ 141.89§	417.28 $\pm$ 119.90	0.175
Vertical power of CG in the second pull (W)	647.57 $\pm$ 239.29	598.71 $\pm$ 213.94	558.28 $\pm$ 172.81‡	0.326#
Absolute total power (W)	1,722.57 $\pm$ 441.44†	1,959.28 $\pm$ 422.40	1,933.42 $\pm$ 392.29	0.345**
Relative total power (W·kg ⁻¹)	22.28 $\pm$ 2.13	25.85 $\pm$ 3.07	25.28 $\pm$ 2.98	0.446
Absolute total power in the second pull (W)	2,844.57 $\pm$ 727.61†	3,108.28 $\pm$ 804.50	3,279.85 $\pm$ 849.77‡	0.262
Relative total power in the second pull (W·kg ⁻¹)	37.28 $\pm$ 4.34	40.28 $\pm$ 4.88	42.71 $\pm$ 3.03‡	0.287

*1RM = 1 repetition maximum.

†Significant difference at the $p < 0.05$ level in 80%1RM.

‡Significant difference at the $p < 0.05$ level in 60%1RM.

§Significant difference at the $p < 0.05$ level in 100%1RM.

|| $p < 0.05$; ¶ $p = 0.051$; # $p = 0.061$; ** $p = 0.070$.

showed that the absolute total power of 60%1RM was significantly higher than that of 80%1RM ($p < 0.05$). As for the relative total power values, however, no significant difference was observed. On the other hand, in the second pull, significant differences were found between the absolute total power values and between the relative total power values of the different percentages of 1RM (Table 5).

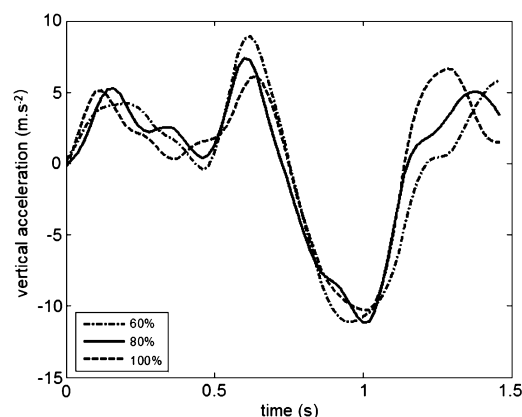


Figure 6. The linear vertical acceleration of the barbell in the snatch lifts of TS: from lift-off to the catch phase.

The analysis of linear vertical acceleration during snatch lifts, an indicator of the magnitude of the force applied on the barbell, revealed 2 peak acceleration values during the first pull of 60%1RM, whereas only 1 peak was seen during the first pull of the other 2 lifts. When the transition phases of the 3 lifts were investigated, it was found that the acceleration of the barbell decreased sharply in 60%1RM snatch lift, whereas in the other 2 lifts, it decreased relatively mildly as the weight of the barbell was increased. It was also observed that the barbell was accelerated positively again in vertical direction during the second pull, 100%1RM being the highest and 80%1RM the lowest. Another finding was that the barbell was accelerated negatively during the catch phase and the magnitude of acceleration of the barbell that started to fall from the maximum height was higher at 60%1RM when compared with 80%1RM or 100%1RM (Figure 6).

DISCUSSION

The most distinctive effect of the increased barbell weight in snatch lifts in this study was the decrease in the velocity of the barbell and the vertical displacement of the barbell and CG and the increase in the vertical velocity of CG during the turnover under the barbell. It was reported in the studies of Garhammer (9,11,14) that when the barbell weight was increased during competition, maximum vertical displacement values of the barbell were decreased. In addition, it was

found in one study (17) that maximum vertical displacement of the barbell during snatch lifts of elite weightlifters was 1.25 m, although this value was 1.15 m in another study carried out by the same authors (15). In this study, the maximum vertical displacement of the barbell at 100%1RM was found to be 1.18 m. The main reason for the inconsistent results found by different researchers about the maximum height of the barbell could be the physical differences of the weightlifters, including the differences in their height.

The velocity-time relationship of the barbell during the snatch lift—especially the maximum vertical velocity of the barbell—is an important point for both coaches and athletes (21). Baumann et al. (2) reported 2 types of velocity curves for the snatch lift of elite weightlifters: The first was with 2 maximum peaks in the vertical velocity during the pull, and the second was with 1 maximum peak. In this study, a slight decrease was observed in vertical velocity during the transition phase in successful lifts, suggesting a curve with almost 2 peaks. Although 1 maximum peak in the vertical linear velocity of the barbell during the pull was reported to be characteristic of better weightlifters, successful lifts by elite weightlifters with 2 maximum peaks were also reported (2). The deceleration of the barbell during the transition phase were characterized by fatigue or higher percentages of their maximum velocities at the end of the first pull and might not cause any notable decrement in performance as long as the loss of the barbell's linear vertical velocity is a small percentage of maximum linear vertical velocity (18).

Although the maximum vertical velocity of the barbell increased in parallel to the increase of body weight in categories (2), it was observed to decrease in 1RM in general (9,10). Therefore, it might be said that increasing the barbell weight is a factor affecting the vertical velocity of the barbell directly. Isaka et al. (21) reported similar results to those of this study regarding the horizontal and vertical displacement of the barbell. Small differences were seen among the values resulted from the physical and technical characteristics of the athletes. The study conducted by Hoover et al. (20) on women weightlifters, and others, support the finding of this study that the maximum vertical velocity of the barbell decreased as the weight of the barbell was increased.

When the body and the barbell are evaluated together as a system, the height and vertical velocity of the barbell decrease as the barbell weight increases, during which the kinematics of the body that constitute the system also change. The most significant indication of this change is the vertical velocity of CG during the turnover under the barbell. When the kinematic characteristics of the system were investigated, it was observed that the full extension of the body was realized at the end of the second pull. At this moment, the CG reached the maximum height and the barbell reached its maximum velocity (5). After this moment, the turnover under the barbell started as the second pull phase ended. Although the barbell continued its upward motion during the turnover under the barbell, the turnover under the barbell was realized

in the direction of CG. The decrease observed at the height of the barbell and at its maximum vertical linear velocity during snatch lifts increased the velocity of the turnover under the barbell of the body and decreased the duration of the turnover under the barbell (Table 3). The most important advantage of the increase in the vertical linear velocity of CG during the turnover under the barbell was the decrease in the drop distance of the barbell; technically, moving faster during the turnover under the barbell resulted in an easier control of the barbell weight, namely, catching the barbell.

The most notable effect of the increase in the barbell weight was that, although it resulted in an increase in the vertical work done during the first pull, it did not produce a similar result in the second pull (Table 4). The increase in the barbell weight also led to an increase in total work values, similarly to those of the first pull. The reason for the increase in the vertical work and total work values in parallel to the increase in the barbell weight during the first pull might be the longer time period needed to overcome the inertia of the barbell and the relatively decreased vertical velocity (18).

Gourgoulis et al. (15,18) reported that the work values of the second pull had an average 409 J value, whereas Gourgoulis et al. (17) found that this value was 388 J. The value obtained in our study was 511 J for 80%1RM and 100%1RM. The biggest reason for the difference between the result obtained in our study and those reported in previous studies might be that Gourgoulis et al. (15,17) did not include the work done by CG in the work calculations, which, therefore, led to lower values in the total work. The second pull work values found in the study of Harbili and Aritan (19) and those of this study, both of which included weightlifters with similar physical characteristics who lifted similar weights, were similar to each other.

It was emphasized that in addition to the vertical movement of the barbell, the horizontal movement was also an important factor in performing the snatch technique (23) because the horizontal displacement of the barbell during snatch lift caused an additional acceleration and work during the lift (2). In this study, it was observed that the horizontal work values between the different percentages of the lifts were similar to each other because there was no statistically significant difference between the total horizontal work values. However, technical efficiency is a parameter that supports the importance of the decrease in the horizontal work. Therefore, it was seen that technical efficiency, which showed the effect of increased barbell weight on the barbell kinematics for snatch lifts, was higher at 100%1RM when compared with 60%1RM and 80%1RM. The values found demonstrated that to make a more efficient lift, athletes presumably increased the vertical work done on the barbell and decreased the horizontal displacement of the barbell.

Garhammer (11) used the change of mechanical energy, which expressed the total of potential and kinetic energy, to calculate the work done and expressed that at the end of the second pull, CG reached the maximum point in the vertical

direction and that at this moment, the kinetic energy of CG was negligible because its velocity was close to zero. Therefore, he proposed the use of potential energy to calculate only the work done by CG in the vertical axis. It was found in this study that as the weight of the barbell increased, there were decreases in the vertical work of CG in the second pull and in the total vertical work (Table 4). The decrease in the vertical work done by CG was directly related to the drop observed in the height of CG, similarly to the decrease observed in the vertical kinematics of the barbell. However, the increase in the work done on the barbell and the decrease in the work done by CG was in accord. The analysis of the decreasing vertical displacement of the barbell and the change in mechanical energy because of velocity indicated that the only factor that supported the increase in the work done on the barbell was the mass of the barbell. The most important variable here was the contribution of the kinetic energy of the barbell to the vertical work done on the barbell. These findings suggested that weightlifters applied more force to the barbell in the vertical direction for a successful lift, and the decrease in the horizontal work done on the barbell supported this conclusion.

The pull refers to the initial part of the lift during which the barbell is displaced from the floor to approximately waist height (5). In a previous study, biomechanical analyses were performed by dividing the pull stage into 3 phases (2,5,7): The first pull, the transition, and the second pull. The pull, with respect to angular displacement at the knee joint, consists of 2 stages of extension, separated by a period of flexion, known as the double knee bend (5). This knee flexion is used to realign the lifter, relative to the barbell, and is referred to as the second knee bend (5). These angular changes in the movement structure of the knee joints made considerable contributions to defining this second pull as the power phase in that these changes transfer the elastic energy stored in knee extensor muscles into the second pull after the knee flexion during transition phase, which was indicative of the biomechanical importance of the double knee bend (2,17). In addition, during the first pull, the changes in the barbell's kinetic and potential energy were greater, and the lifters had to produce considerable work over a long period to overcome the inertia of the barbell (18). In this study, it was found that as the barbell weight increased, the duration of the first pull extended. This finding supported the result reported by Gourgoulis et al. (17) that the first pull was forceful in nature, whereas the second pull was characterized by power.

According to Garhammer (12), even a small decrease of about 5% in the barbell weight led to an increase in the movement velocity, which might increase the power output substantially as a result. It was found in the same study that as the barbell weight increased, power outputs during the first and second pulls decreased. Therefore, the findings of Garhammer (12) seemed to be inconsistent with the significant increase observed in the first and second pull power values of the different percentages of 1RM in this study.

This difference between the 2 studies might have resulted from the fact that unlike the greater difference between the percentages of 1RM in this study, the percentages of 1RM in the study of Garhammer (12) were performed with barbell weights close to 1RM (92.7, 97.6, and 100%). This closeness in the percentages of 1RM probably forced the limits of the weightlifters' performance.

The mean power output value during the second pull was reported as 2,506 W in a study performed by Gourgoulis et al. (17). The average value found in our study, however, was 3,279 W. The reason for this difference was probably because Gourgoulis et al. (17) calculated only the vertical power during the second pull. It was reported in another study that as the body weight increased, so did their second pull power values; as the bar weight was increased, the duration of the lifts extended, and the average vertical velocity, maximum bar weight, and power outputs decreased (9).

As the barbell weight was increased, the vertical and horizontal kinematics of the body and barbell decreased during the first and second pulls in this study. The decrease in the maximum height and vertical velocity of the barbell observed during the total pull was found to be characterized by the barbell's decreased and the body's increased kinematics during the turnover under the barbell. With the effect of the increased barbell weight, the values of the work done and power output increased during the first pull, but during the second pull, power output increased, whereas the work done remained unchanged.

PRACTICAL APPLICATIONS

The rationale for this study was that analyzing the effects of increased barbell weights on the barbell and body kinematics of elite weightlifters at different percentages of 1RM (60, 80, and 100%) would facilitate the understanding of the biomechanical demands at each phase in the snatch lift.

The results of this study are consistent with those of previous research studies reporting that the strength-oriented first pull and the power-oriented second pull are the most important phases in the snatch technique and that the explosive strength during the second pull is especially important for a successful turnover under the barbell. The decrease in horizontal power despite the increases in vertical power observed in our study indicates that coaches and practitioners should primarily focus on designing training programs for the improvement of the explosive strength for a better pull. Based on the result that the power output in 80 and 100% of 1RM were similar, it is recommended that loads between submaximal and maximal barbell weights (80 and 100% of 1RM) should be chosen in training programs designed for improving strength, preferably loads of 100% of 1RM.

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